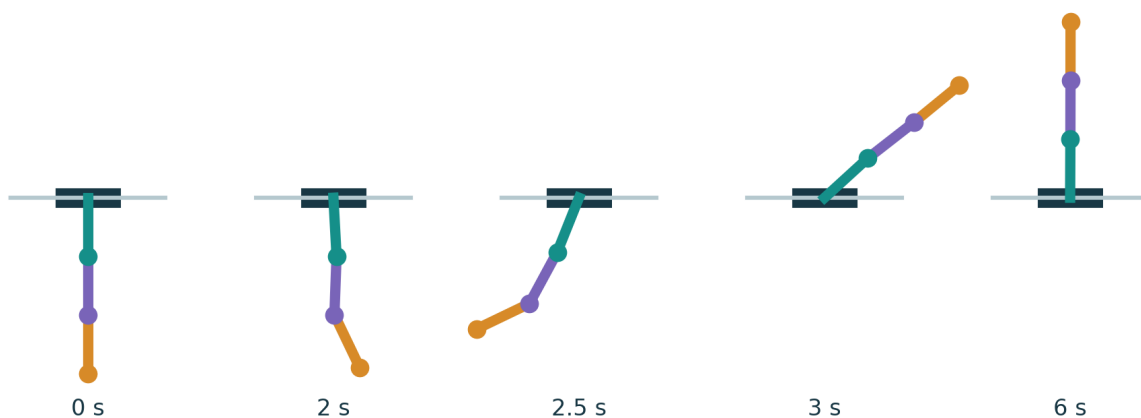


From one link to three

Modelling, feedback control and nonlinear swing-up of a cart-mounted pendulum

Can one horizontal force stabilise three freely rotating links?

This project develops a Python simulation and an interactive browser demonstration, progressing from a single inverted pendulum to a coupled three-link system. The emphasis is on connecting equations, controller design and observable behaviour.



Five snapshots from the computed six-second swing-up. Cart centres are aligned for comparison; horizontal cart travel is shown quantitatively on page 6. Colours identify the three links.

What the project demonstrates

Model: eight state variables and one cart actuator, with nonlinear inertial coupling, gravity and force saturation.

Design: local LQR stabilisation, an offline swing-up reference and time-varying feedback along that reference.

Verification: equation checks, closed-loop eigenvalues, energy accounting and comparison of successful and failed trials.

Connection to Control Principles / MECH3610

The work applies state-space modelling, linearisation, controllability, state feedback and the distinction between local stability and constrained nonlinear behaviour. Earlier single-link experiments also compare pole placement with LQR. This note focuses on the more demanding three-link calculation.

This project explores pendulum control through simulations based on an idealised physical model. In a real setup, additional effects such as joint friction, sensor noise and motor limitations could affect performance and would need to be taken into account.

AI assistance: AI tools were used to assist with code development, debugging and preparation of this report. The code and saved simulation results provide a reproducible record of the calculations.

01 / Derive the physical model

Rigid links, free joints and coupled equations of motion

Angles are measured from the upward vertical. Coordinates and state are defined as follows; the state interleaves positions and velocities.

$$q = [x, \theta_1, \theta_2, \theta_3]^T, \quad s = [x, v, \theta_1, \omega_1, \theta_2, \omega_2, \theta_3, \omega_3]^T$$

Cart mass $M_c = 0.8$ kg; each point mass $m_i = 0.2/3$ kg; each massless rod $l_i = 0.5/3$ m. Gravity $g = 9.81$ m/s²; cart friction = 0.08 N·s/m. Define W_i as the sum of masses from joint i to the final mass. Indices i, j run from 1 to 3; index 0 denotes the cart.

1. Geometry and energy

$$X_i = x + \sum_{k=1}^i l_k \sin \theta_k, \quad Y_i = \sum_{k=1}^i l_k \cos \theta_k$$

$$T = \frac{1}{2} \dot{q}^T M(q) \dot{q}, \quad V = g \sum_i W_i l_i \cos \theta_i, \quad \mathcal{L} = T - V$$

2. Lagrange equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = Q_{\text{ext}}, \quad M(q) \ddot{q} = b(q, \dot{q}, u)$$

$$M_{00} = M_c + \sum_i m_i, \quad M_{0i} = W_i l_i \cos \theta_i, \quad M_{ij} = W_{\max(i,j)} l_i l_j \cos(\theta_i - \theta_j)$$

The off-diagonal mass-matrix entries couple cart acceleration and joint accelerations. Moving one mass therefore affects the other links without adding joint motors.

3. Forces and numerical acceleration

$$b_0 = u - 0.08 \dot{x} + \sum_i W_i l_i \sin \theta_i \dot{\theta}_i^2$$

$$b_i = g W_i l_i \sin \theta_i - \sum_j W_{\max(i,j)} l_i l_j \sin(\theta_i - \theta_j) \dot{\theta}_j^2$$

At each integration stage, solve the 4×4 linear system for accelerations, then advance all eight states together. The angle-dependent matrix and velocity-squared terms remain in the simulation; they are not replaced by the linear approximation.

The complete mass matrix

How cart motion and the three joint accelerations are coupled

Rows and columns follow the coordinate order: cart position, then the three absolute link angles. The matrix is symmetric. Off-diagonal entries are highlighted in teal.

$$M(q) = \begin{bmatrix} M_c + W_1 & W_1 l_1 \cos \theta_1 & W_2 l_2 \cos \theta_2 & W_3 l_3 \cos \theta_3 \\ W_1 l_1 \cos \theta_1 & W_1 l_1^2 & W_2 l_1 l_2 \cos(\theta_1 - \theta_2) & W_3 l_1 l_3 \cos(\theta_1 - \theta_3) \\ W_2 l_2 \cos \theta_2 & W_2 l_1 l_2 \cos(\theta_1 - \theta_2) & W_2 l_2^2 & W_3 l_2 l_3 \cos(\theta_2 - \theta_3) \\ W_3 l_3 \cos \theta_3 & W_3 l_1 l_3 \cos(\theta_1 - \theta_3) & W_3 l_2 l_3 \cos(\theta_2 - \theta_3) & W_3 l_3^2 \end{bmatrix}$$

$$W_1 = m_1 + m_2 + m_3, \quad W_2 = m_2 + m_3, \quad W_3 = m_3$$

Here M_c is the cart mass, m_i is the point mass at the end of link i , and l_i is the length of link i . Each W_i includes that endpoint and every mass downstream of it.

Read the first row as one equation

$$(M_c + W_1)\ddot{x} + W_1 l_1 \cos \theta_1 \ddot{\theta}_1 + W_2 l_2 \cos \theta_2 \ddot{\theta}_2 + W_3 l_3 \cos \theta_3 \ddot{\theta}_3 = b_0$$

The first row is the cart equation. It contains the angular accelerations of all three links as well as the cart acceleration. The right-hand side b_0 , defined on the preceding page, contains applied force, cart friction and velocity-dependent terms.

One coupling term, interpreted physically

$$M_{02} = M_{20} = W_2 l_2 \cos \theta_2$$

This entry multiplies the angular acceleration of the second link in the cart equation. Its symmetric counterpart multiplies cart acceleration in the second-link equation. Coupling depends on the instantaneous configuration through the cosine factor.

The off-diagonal mass-matrix entries couple cart acceleration and joint accelerations. Moving one mass therefore affects the other links without adding joint motors. Gravity and velocity-dependent terms also contribute to the complete motion.

Why the full matrix matters

The simulator solves the coupled 4×4 system at each integration stage. Integrating each bar as an independent pendulum would discard these coupling terms and describe a different physical system.

02 / Calculate the controller

Local stabilisation and a separate strategy for getting upright

Linearisation and controllability

$$\dot{s} = As + Bu, \quad J = \int_0^{\infty} (s^T Q s + R u^2) dt$$

The matrices A and B are obtained by linearising the dynamics about the upright equilibrium. The controllability matrix $[B, AB, A^2B, \dots, A^7B]$ has numerical rank **8**, equal to the number of state variables. Thus, despite having only one actuator, the linearised system is fully controllable: a suitably chosen cart force can influence all eight state components over time through mechanical coupling. This supports the use of state-feedback control, but does not guarantee recovery from large disturbances in the nonlinear system with the ± 8 N actuator limit.

Why solve the Riccati equation?

Solving this equation finds the stabilising matrix P, from which the feedback gain K is calculated. The resulting control minimises J for the unconstrained linear model, balancing deviations from equilibrium against actuator effort according to Q and R. This gives a systematic way to choose the feedback coefficients rather than tuning each one separately.

$$A^T P + PA - PBR^{-1}B^T P + Q = 0, \quad K = R^{-1}B^T P$$

$$u = \text{clip}(-Ks, -8, 8), \quad \dot{s}_{\text{linear}} = (A - BK)s$$

$$Q = \text{diag}(10, 1, 100, 1, 100, 1, 100, 1), \quad R = 0.1$$

These weights use SI state coordinates. Larger angle penalties prioritise staying upright; changing units requires transforming the weights.

Computed gain (rounded, in the state order above):

$$K = [-10.000, -15.837, -544.931, -15.082, 1639.841, 75.488, -1221.599, -101.599]$$

Check the computed controller

Two distinct checks: closed-loop stability and numerical solution accuracy

1. Do small perturbations decay?

Eigenvalues of $A - BK$: -30.586; $-16.477 \pm 4.017i$;
 $-8.889 \pm 4.457i$; -10.097; $-1.001 \pm 0.860i$ (s^{-1}).

$$\lambda = a + ib, \quad |e^{\lambda t}| = e^{at} \rightarrow 0 \quad \text{when } a < 0$$

All eigenvalues have negative real parts. Each linear mode therefore decays with time; a nonzero imaginary part adds an oscillation. This confirms that the computed gain K asymptotically stabilises the upright equilibrium of the linearised model. It does not guarantee recovery after large disturbances when nonlinear dynamics and the force limit matter.

2. Does the computed P satisfy Riccati?

$$E = A^T P + PA - PBR^{-1}B^T P + Q, \quad \frac{\|E\|_F}{\|Q\|_F} = 2.33 \times 10^{-10}$$

Substituting the computed P into the Riccati equation should give zero. The residual E measures the remaining numerical discrepancy. The Frobenius norm combines all matrix entries into one magnitude; dividing by the norm of Q expresses it relative to the chosen state-cost scale.

The relative residual is only 2.33×10^{-10} . Thus, the computed P satisfies the Riccati equation to high numerical accuracy. Together with the stable closed-loop eigenvalues, this supports using the resulting K in the nonlinear simulation. The residual is a solver check, not an error in pendulum position or a guarantee of robustness.

Why swing-up needs another calculation

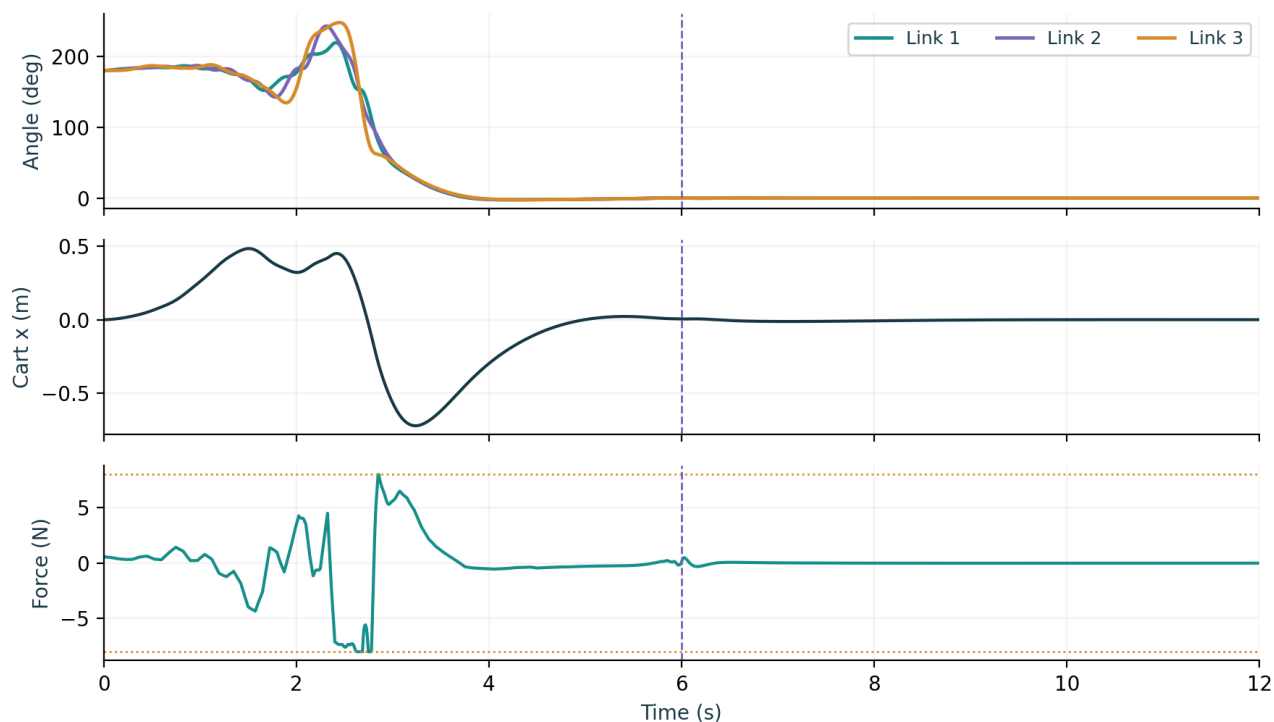
$$u(t) = \text{clip}[u_{\text{ref}}(t) - K(t)(s - s_{\text{ref}}(t)), -8, 8]$$

Starting downward is far outside the local upright design. A six-second reference is computed offline using Hermite-Simpson collocation. The nonlinear equations are then linearised along it; a backward discrete Riccati recursion supplies $K(t)$ at 0.01 s intervals. At 6 s, control switches to the upright LQR. Independent nonlinear integration checks the resulting motion.

The trajectory is a feasible numerical demonstration, not a proof of global optimality. Clipping the force also removes the unconstrained LQR optimality guarantee.

03 / Compare prediction and outcome

Saved nonlinear trials with the same three-link model and ± 8 N actuator



Downward start: all angles initially 180° . Dashed line: handoff to upright LQR at 6 s. Peak cart displacement is approximately 0.72 m. Rapid early rotations are part of this computed manoeuvre.

Trial	Settling time*	Peak force	Outcome
Aligned $+1^\circ / +1^\circ / +1^\circ$	2.133 s	2.211 N	Settles
Opposed $+1^\circ / -1^\circ / +1^\circ$	—	8.000 N	Cutoff at 0.295 s
3 N push, 4.0-4.2 s	6.323 s	4.183 N	Settles
Downward swing-up	6.500 s	8.000 N	Settles

*Times are measured from simulation start. The push trial settles 2.123 s after the disturbance ends. Settling requires $|x| \leq 0.02$ m, $|v| \leq 0.03$ m/s, all angles $\leq 1^\circ$ and angular speeds ≤ 0.05 rad/s for the remaining run, with at least 1 s remaining.

The most informative result is also a failure

Opposing initial angles demand more control authority than the saturated local controller can supply. Even 1° per link can fail in this direction of the state space. Negative linear closed-loop poles therefore do not imply that every small-looking nonlinear trial will recover.

04 / Verification and next questions

What the demonstration supports, and what remains to be established

Interaction must obey the same mechanics

$$Q_{\text{mouse}} = J_p(q)^T F_{\text{mouse}}, \quad \dot{E} = u\dot{x} - 0.08\dot{x}^2 + F_{\text{mouse}}^T \dot{p}$$

Dragging a point applies a bounded spring-damper force through the point-position Jacobian. It does not set joint angles. The power identity above checks how cart actuation, friction and mouse work change mechanical energy. The three hinges remain unactuated.

For strong disturbances, the interactive controller attempts to settle downward before relaunching swing-up. Some tested pulls recover; arbitrary recovery is not guaranteed. These live trajectories are separate from the saved benchmark results on page 6.

Checks behind the calculations

Model consistency: equilibrium, finite-difference linearisation and an independently assembled mass-matrix/Jacobian calculation.

Controller consistency: Riccati residual, controllability and closed-loop eigenvalues.

Numerics: nonlinear Runge-Kutta integration; saved triple-link step 0.0025 s, live step 0.00125 s. Display interpolation preserves fixed rod lengths.

Experimental discipline: record force saturation and failures alongside successful trials; preserve CSV traces and calculation scripts.

Limitations worth discussing

The model assumes point masses, massless rigid rods, frictionless hinges, perfect state measurements and an ideal force actuator. It does not include motor bandwidth, sensor delay, joint friction, contact or hardware identification. The 2 m travel stop and 60° local-trial cutoff are simulation limits, not collision models.

Next scientific step

Map the region of successful recovery over initial angles and velocities. Then add sensor noise and partial observation, design a state observer, and measure how delay and force limits change the success region. These extensions would test robustness rather than only produce a more elaborate animation.

Reproduce and present

Open [visualization/triple.html](#) for the offline interactive demo. Run `python src/build_triple_demo.py` in the project environment to rebuild the saved trials. The calculation trail is in `src/triple_pendulum.py`, `src/triple_swingup.py`, `src/plan_triple_swingup.py` and `results/triple/`. The README and RESEARCH_LOG document the progression and revisions.

Suggested demonstration: show the downward swing-up, inspect the force limit, contrast aligned and opposed initial angles, then explain the equations and why a local controller can fail. This PDF is self-contained; share the project folder separately if the professor wants to run the demo.

From massless links to massive rods

Reading guide: pages 1–7 describe the original massless-rod model and its controller. This extension adds uniformly distributed rod mass; its new results must not be confused with the earlier benchmark.

Physical change. Each of the three rigid rods has mass 0.010 kg and length 1/6 m. Endpoint masses remain 0.2/3 kg each. Suspended mass increases from 0.200 to 0.230 kg. The cart remains 0.8 kg and the motor force limit remains ± 8 N.

New energy terms. Each rod contributes translation of its centre of mass, rotation about its centre and gravitational potential energy. For a uniform rod, $I = mL^2/12$. These terms modify the coupled mass matrix and gravity terms; adding rod mass only to the cart would be incorrect.

Control recalculation. The upright linearisation, Riccati solution and feedback gains are recalculated using the same Q and R . A six-second swing-up reference is re-optimised with the massive-rod equations. The old trajectory is only an initial guess. Time-varying LQR tracks the new reference, then upright LQR balances the system.

Measured forward replay. Starting exactly downward and at rest, the simulated pendulum reaches the specified settling region at **6.38 s**. Peak force is **8.00 N**; maximum absolute cart displacement is **0.721 m**. The simulation continues to 12 s.

Verification. Repeating at 0.000625 s instead of 0.00125 s gives the same settling time. Maximum angle discrepancy is below 6.3×10^{-7} rad. Independent energy calculations, power balance, the zero-rod-mass limit and finite-difference linearisation are tested. These are numerical checks, not hardware uncertainty bounds.

The optimizer reached its 600-evaluation budget. No global or time-optimal solution is claimed. The independent nonlinear replay, including force saturation, is the acceptance check.

Limitations and future work

Completed: massive-rod near-upright experiments; a recomputed nominal swing-up; an offline animation with trajectory trails, force readout and slow playback. The original massless-rod demo separately includes mouse interaction.

Known failure: opposing initial angles ($+1^\circ$, -1° , $+1^\circ$) reach the tilt cutoff under the local saturated LQR. Successful aligned-tilt and swing-up trials do not imply recovery from every state.

Physical scope: rigid uniform rods, frictionless joints, ideal force actuation and perfect state measurements. Cart viscous friction is included. Motor dynamics, joint friction, backlash, collisions and measurement delay are not yet represented. A real apparatus could behave differently.

Future work — not yet implemented

1. **Robustness study:** vary masses, initial states and disturbance timing without retuning; report successes and failures under fixed criteria.
2. **Imperfect sensing:** include measurement noise, delays and a state observer.
3. **Massive-rod interaction:** adapt external mouse forces and recovery to this model, preserving mechanical consistency.
4. **Alternative swing-up:** compare the planned trajectory with online replanning or model predictive control. Energy-based swing-up is another exploratory direction; transferring it to three links is not automatic.
5. **Hardware validation:** identify parameters and actuator limits, then compare measured trajectories with the model.

Scientific question for the next stage: when is a precomputed trajectory with feedback sufficient, and when does replanning improve recovery from disturbances?

AI assistance and authorship. This is Louis Gosset's educational project. AI tools assisted code development, debugging, mathematical exposition and report preparation. Implemented experiments, numerical checks and proposed work are distinguished explicitly. This report does not claim experimental hardware validation or independent invention of established control methods.

Reproduction: see the repository README and docs/massive-rods.md. Reference trajectories, CSV results and validation metrics are included. The source, calculations and limitations are intended to support technical discussion.